Silas Bambil

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Education

Georgia State University 2026–Present
MS Mathematics, concentration in Statistics United States

Georgia State University 2025
MS Quantitative Risk Analysis and Management United States

Kwame Nkrumah University of Science and Technology 2020
BSc Actuarial Science Ghana

Research Interests

Statistical methods for finance; credit-risk model updating; distribution shift; model selection and validation; extreme value theory; tail-risk measurement; statistical inference and simulation.

Research Experience

Tail Model Diagnostics Using Expected Shortfall and Bounded Loss Layers with an Application to Fire Losses 2026
Manuscript in preparation
Silas Bambil, Shudeshna Nandy, Clement Ampong, Prakash Karki, and Chi-Kuang Yeh

- Developed bounded-layer diagnostics to assess discrepancies between empirical loss distributions and fitted tail models beyond high-loss quantiles.
- Formulated discrepancy measures using established stop-loss identities and evaluated inferential procedures through simulations assessing detection, localization, and interval coverage.
- Applied the methods to 2016–2025 U.S. structure-fire loss records to examine how exceptional observations affect Expected Shortfall and bounded-layer estimates.

Selection Regret in Credit-Risk Model Updating Under Distribution Shift: The Role of Recent Labeled Information 2026
Independent manuscript in preparation
Silas Bambil

- Studied predictive regret from selecting among fitted credit-risk model-updating policies under distribution shift.
- Distinguished how often the best fitted candidate is selected from the predictive cost of selecting an alternative policy.
- Used Monte Carlo experiments and independent test samples to examine how recent-label sample size affects selection and regret.

Selecting and Estimating Credit-Risk Model Updates Under Distribution Shift 2026
Manuscript in preparation
Silas Bambil and Shudeshna Nandy

- Compared model retention, intercept recalibration, intercept-and-slope recalibration, and refitting under changes in the distribution of credit-risk data.
- Evaluated validation-guided policy selection using simulation experiments with separate training, validation, and test samples.
- Examined how recent labeled observations affect model updating and predictive performance, including an application to historical mortgage-credit data.

Graduate Admissions Predictive Analysis 2024
Graduate research project, Georgia State University United States

- Prepared applicant-level datasets and constructed predictors for graduate enrollment modeling.
- Supported evaluation of enrollment prediction models and interpretation of applicant-level predictors.

Application of Extreme Value Theory to Fire Insurance 2020
Undergraduate thesis, Kwame Nkrumah University of Science and Technology Ghana

- Applied extreme value methods to model severe fire-insurance losses and examine estimates of tail risk.

Academic Appointments

Georgia State University 2023–2025
Graduate Assistant United States

- Consolidated graduate admissions records and constructed applicant-level predictors for enrollment modeling.
- Supported statistical analysis and evaluation of predictive models for faculty research.

Teaching Experience

Georgia State University 2026–Present
Graduate Lab Assistant United States

- Provide instructional support for College Algebra (MATH 1111) and Support for College Algebra (MATH 0999).
- Assist students with algebraic methods, mathematical reasoning, and problem solving.

Osceola School for the Arts 2026
AP Mathematics Teacher United States

- Taught AP Calculus, AP Precalculus, and AP Statistics.
- Developed lessons and assessments to evaluate student understanding.
- Reported a 70% AP examination pass rate among students taught.

Professional Experience

OPH CHEM CONSULT 2022–2023
Finance Associate Ghana

- Examined financial misappropriation risks and community financial challenges.
- Summarized findings in analytical reports and research briefs.

Ghana Statistical Service 2020–2021
Research Assistant Ghana

- Supported survey research and management of population and economic data.
- Contributed to statistical analysis and preparation of research reports.

Teaching and Research Skills

Teaching: Lesson planning, classroom management, assessment design, mathematical instruction, student mentoring, and learning evaluation.
Research: Research methodology and framework development, simulation design, statistical model validation, Monte Carlo simulation, bootstrap inference, sensitivity analysis, reproducible computation, and academic writing.

Technical Skills

Software: R, Python, Microsoft Excel, Microsoft Office.
Statistical Methods: Regression analysis, Monte Carlo simulation, bootstrap methods, cross-validation, model calibration and validation, statistical visualization.
Risk Modeling: Extreme value theory, peaks-over-threshold methods, generalized Pareto distributions, Value-at-Risk, Expected Shortfall, and credit-risk modeling.

Awards

Graduate Merit Scholarship, Department of Mathematics and Statistics, Georgia State University 2026
Graduate Merit Scholarship, Robinson College of Business, Georgia State University 2023–2025

FINRA Examinations

Series 63 – Uniform Securities Agent State Law Examination 2025
Securities Industry Essentials (SIE) 2025